Oumayma El Moussaoui

Machine Learning Engineer · Data Scientist

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Portfolio: elmoussaouioumayma.github.io/Portfolio
GitHub: github.com/elmoussaouioumayma

PROFILE

ML engineer with an M.Sc. in Machine Learning & Data Analytics and an engineering background. I build machine-learning and computer-vision systems end to end, from data and model to validation, deployment as a working application and hand-over to non-technical users. My focus is on results that can be trusted in production: leakage-free evaluation, grouped cross-validation, testing against known ground truth, and metrics chosen for the real use case (e.g. recall-oriented thresholds in health screening). I enjoy working with domain experts and taking ownership of a system from first idea to the version people rely on.

EXPERIENCE

Academic Employee – Machine Learning & Image Analysis Nov 2024 – present
Zentrum Elektrochemische Oberflächentechnik (ZEO), Hochschule Aalen · Wissenschaftliche Mitarbeiterin since Feb 2026, previously Student Research Assistant

- Own the image-analysis software that automatically detects, classifies and measures cracks in microscope images of chromium coatings, from requirements to the tools researchers use every day.
- Replaced manual annotation (2–2.5 h per image) with an automated, validated Python pipeline; chamfer distance to manual ground truth < 0.08 µm.
- Built self-service tools for researchers: live parameter tuning, batch processing with CSV export, depth and width profiles along cracks.
- Analyse process and measurement data from electroplating experiments and present results to chemists and engineers.
- Model process knowledge as an ontology (Protégé/OWL) so experimental data can be structured and reused.

Engineering Intern – Energy Efficiency Sep – Nov 2021
Excellence Hub, El Jadida, Morocco

- Energy audit of an industrial plant: collected and analysed consumption data, quantified losses and proposed savings measures.

Engineering Intern – Thermal Processes Apr – Jun 2019
LafargeHolcim, Meknes, Morocco

- Analysed thermal losses of a cement kiln and evaluated insulation options.

Technical Intern – Process Performance Apr – May 2018
OCP Group, Khouribga, Morocco

- Monitored an industrial grinder and proposed cost-effective measures to improve its performance.

PROJECTS · live demos and code on my portfolio and GitHub

Crack Depth Explorer – 3D measurement from four photos 2026 · live web app

- End-to-end computer-vision system: photometric stereo reconstructs the 3D surface from four photos, then every crack is detected and measured in millimetres (depth, width, length).
- Built a synthetic data generator with known 3D ground truth to evaluate the pipeline objectively on 20 unseen scenes (41,299 test points).
- **0.21 mm** mean depth error, **2.6×** more accurate than a single-photo baseline; detection precision / recall **0.96 / 0.97**. Deployed as an in-browser application.

Stack: Python, NumPy/SciPy, OpenCV, scikit-image, JavaScript, Three.js

Crack Network Graph – from pixels to a measured graph 2026 · live web app

- Turns a photo of a cracked surface into a graph (segments = edges, junctions = nodes) with lengths, widths, directions, T / Y / X junction types and closed cells; exports GeoJSON, GraphML and CSV.
- Evaluated on 30 generated images with a known network: junction recall / precision **0.92 / 0.97**, **90 %** junction types correct, formation type recognised in **30 / 30** images.

Stack: Python, OpenCV, scikit-image, NetworkX, Gradio, JavaScript

Dark Pattern Detection in User Interfaces – NLP

- Text classifier for manipulative website content; cross-validation grouped by website to prevent train/test leakage. Gradio app that also reads screenshots via OCR. **97.3 %** test F1.

Stack: Python, scikit-learn, Hugging Face Transformers, TF-IDF, OCR, Gradio

Diabetes Prediction – ML for health screening

- Compared six models in leakage-free pipelines and tuned the decision threshold for screening, where a missed case costs more than a false alarm. **CV AUC 0.84**, **85 %** recall on the test set.

Stack: Python, scikit-learn, XGBoost, pandas

Fake vs. Real News Detection – data quality & NLP

- Found and removed data leakage (5,498 duplicates, source markers revealing the label), then trained TF-IDF + Logistic Regression and tested it on newly scraped articles. **97.1 %** accuracy after cleaning.

Coating Layer Analysis – image analysis

- Detects coating layers in cross-section images, traces their boundaries and measures thickness; layer count chosen with a statistical model (BIC). Web app and batch tool. **95 %** correct layer count, **0.66 µm** median boundary error (100 test images).

SKILLS

Programming	Python (pandas, NumPy, SciPy), SQL, R, JavaScript, Git/GitHub, Jupyter
Machine learning	scikit-learn, XGBoost, NLP (TF-IDF, Hugging Face Transformers), model evaluation, grouped cross-validation, leakage detection, threshold tuning
Computer vision	OpenCV, scikit-image: segmentation, 3D reconstruction (photometric stereo), skeletonization, graph extraction (NetworkX)
ML lifecycle	End-to-end pipelines, synthetic ground truth for validation, batch processing, interactive apps (Gradio, web), reproducible reporting
Data & BI	Power BI, matplotlib, data cleaning, data quality analysis, ontologies (Protégé/OWL)
Engineering	Energy audits, thermal analysis, COMSOL, PVsyst, AutoCAD, SolidWorks

EDUCATION

M.Sc. Machine Learning and Data Analytics 2024 – 2026
Hochschule Aalen, Germany
Thesis: *Optimized Image Processing for Quantitative Crack Analysis in Chromium Layers* – automated pipeline (5 segmentation methods compared, skeleton-to-graph vectorisation, measurement in µm), validated against manual ground truth on 20 images.

Doctoral studies in Energy Engineering (1 year, not continued) 2021 – 2022
ENSIAS, Rabat, Morocco · research on radiative sky cooling with emissive materials

Master's in Energy and Environmental Sciences 2019 – 2021
ENSAM, Rabat, Morocco

Bachelor's in Physical Energy Technologies 2018 – 2019
ENS, Rabat, Morocco

LANGUAGES

Arabic (native) · French (fluent) · English (fluent) · German (B2, telc)

CERTIFICATES

Data Engineer – DataCamp (2023) · Data Analyst – Google (2022)